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Adjustable hinge fasteners for use with a bidet apparatus

Abstract

An adjustable bidet washing apparatus includes two or more adjustable disk hinge fasteners configured for adjustably securing the apparatus, wherein the adjustable disk hinge fasteners each include a disk having two or more openings along its perimeter, and a securing slot for securing the apparatus to a toilet bowl, and a bracket holder having two bracket holder arms, wherein each bracket holder arm includes a pin or a dowel configured to engage one of the two or more openings along the perimeter of the disk; or a bracket holder having two bracket holder arms, wherein each bracket holder arm includes an opening; and a disk having at least two pins or dowels along its perimeter, each configured to engage one of the one or more openings on the two bracket holder arms, and a securing slot for securing the apparatus.

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Background/Summary

TECHNICAL FIELD OF THE INVENTION

(1) The disclosure generally relates to a bidet washing apparatus and more particularly to adjustable hinge fasteners for use with a bidet apparatus.

BACKGROUND OF THE INVENTION

(2) Bidet apparatuses for washing and cleaning body parts (e.g., genital and/or anal regions) with water were originally developed in the form of a bidet that provided a single spray of water from a permanently attached fixture built into a toilet bowl. However, such bidets were expensive and a

new generation of attachable bidets with a plurality of nozzles for multiple water sprays were developed. These bidets may be attached to the sitting area of a toilet bowl for washing the body parts of a person, as well as the bidet itself, using water sprayed from the bidet nozzles.

(3) Various attachable bidet designs have provided some of the desired washing and cleaning effects; however, existing bidets fail to address all concerns related to the design and function of bidets. For example, some attachable bidet apparatuses may not be suitable for use with all toilet bowls due to the structural differences or variations present in the connecting portions of the apparatus and the securing holes present on a toilet bowl. In addition, contamination or pollution of the bidet apparatus during use, including the under surfaces of the bidet and the outer surfaces of the nozzles, is a common problem that causes aesthetic and hygienic issues. Cleaning these surfaces may be problematic due to the inconvenience and difficulty in effectively reaching all of the contaminated areas, especially with the way most bidet apparatuses are more or less permanently attached to the toilet bowl. These problems are particularly important in bidets used, for example, by infirm or sick people who have to be especially cautious about maintaining hygiene and preventing infections.

(4) Thus, there remains a need for an improved attachable bidet apparatus that allows a user to easily attach and/or clean the apparatus and toilet for aesthetic and hygienic reasons.

SUMMARY OF THE INVENTION

(5) The following detailed description is of the best currently contemplated mode of carrying out exemplary embodiments herein. The description is not to be taken in a limiting sense but is made merely for the purpose of illustrating the general principles of the exemplary embodiments herein, since the scope of the disclosure is best defined by the appended claims. Various inventive features are described below that may each be used independently of one another or in combination with other features.

(6) Broadly, embodiments of the disclosure herein generally provide a bidet washing apparatus adjustably attachable to a toilet bowl for cleaning one or more body parts of a user, which includes two or more adjustable disk hinge fasteners configured for adjustably securing the bidet washing apparatus to the toilet bowl, wherein the two or more adjustable disk hinge fasteners each include a disk having two or more openings along its perimeter, and a securing slot for securing the bidet washing apparatus to a toilet bowl; and a bracket holder having two bracket holder arms, wherein each bracket holder arm includes a pin or a dowel configured to engage one of the two or more openings along the perimeter of the disk; or a bracket holder having two bracket holder arms, wherein each bracket holder arm includes an opening; and a disk having at least two pins or dowels along its perimeter, each configured to engage one of the one or more openings on the two bracket holder arms, and a securing slot for securing the bidet washing apparatus to a toilet bowl; one or more water inlets configured to supply water to the bidet washing apparatus; a control unit, housing one or more valves connected to the one or more water inlets, including one or more control switches configured to operate the one or more valves, wherein the one or more valves control water flow from the one or more water inlets; and a nozzle assembly including at least one washing nozzle, connected to at least one of the one or more valves with one or more water tubes, wherein the at least one washing nozzle is positionable for directing water to the one or more body parts of the user.

(7) In one aspect, the disclosure provides a bidet washing apparatus, wherein the disks are polygon-shaped disks.

(8) In another aspect, the disclosure provides a bidet washing apparatus, wherein the disks are each independently a geometric shaped disk selected from a circle, triangle, rectangle, rhombus, square, and a trapezoid.

(9) In another aspect, the disclosure provides a bidet washing apparatus, wherein the securing slot on the disks is a U-shaped securing slot.

(10) In another aspect, the disclosure provides a bidet washing apparatus, wherein the pin or dowel

on each of the two bracket holder arms is inserted into one of the two or more openings along the perimeter of the disk at about 180 degrees to each other.

(11) In another aspect, the disclosure provides a bidet washing apparatus, wherein the at least two pins or dowels along the perimeter of the disk are inserted into the opening on the bracket holder arms at about 180 degrees to each other.

(12) In another aspect, the disclosure provides a bidet washing apparatus, wherein each bracket holder can rotate vertically about the pins or dowels of the bracket holder arms or the pins or dowels of the disks.

(13) In another aspect, the disclosure provides a bidet washing apparatus, wherein the securing slots on the disks are substantially parallel to each other.

(14) In another aspect, the disclosure provides a bidet washing apparatus, wherein the securing slots on the disks can each independently rotate about a vertical axis from about -45 degrees to about +45 degrees.

(15) In another embodiment, the disclosure provides a bidet washing apparatus adjustably attachable to a toilet bowl for cleaning one or more body parts of a user, which includes two or more adjustable disk hinge fasteners configured for adjustably securing the bidet washing apparatus to the toilet bowl, wherein the two or more adjustable disk hinge fasteners each includes a disk having a securing slot for securing the bidet washing apparatus to a toilet bowl; a half ring having two or more openings along its perimeter, and an internal ridge, lip or slot for securing the disk; and a bracket holder having two bracket holder arms, wherein each bracket holder arm includes a pin or a dowel configured to engage one of the two or more openings along the perimeter of the half ring; or a disk having a securing slot for securing the bidet washing apparatus to a toilet bowl; a bracket holder having two bracket holder arms, wherein each bracket holder arm includes an opening; and a half ring having at least two pins or dowels along its perimeter, each configured to engage one of the one or more openings on the bracket holder arms, and an internal ridge, lip or slot for securing the disk; one or more water inlets configured to supply water to the bidet washing apparatus; a control unit, housing one or more valves connected to the one or more water inlets, including one or more control switches configured to operate the one or more valves, wherein the one or more valves control water flow from the one or more water inlets; and a nozzle assembly including at least one washing nozzle, connected to at least one of the one or more valves with one or more water tubes, wherein the at least one washing nozzle is positionable for directing water to the one or more body parts of the user.

(16) In one aspect, the disclosure provides a bidet washing apparatus, wherein the disks are polygon-shaped disks.

(17) In another aspect, the disclosure provides a bidet washing apparatus, wherein the disks are each independently a geometric shaped disk selected from a circle, triangle, rectangle, rhombus, square, and a trapezoid.

(18) In another aspect, the disclosure provides a bidet washing apparatus, wherein the securing slot on the disks is a U-shaped securing slot.

(19) In another aspect, the disclosure provides a bidet washing apparatus, wherein the pin or dowel on each of the two bracket holder arms is inserted into one of the two or more openings along the perimeter of the half ring at about 180 degrees to each other.

(20) In another aspect, the disclosure provides a bidet washing apparatus, wherein the at least two pins or dowels along the perimeter of the half ring are inserted into the opening on the bracket holder arms at about 180 degrees to each other.

(21) In another aspect, the disclosure provides a bidet washing apparatus, wherein each bracket holder can rotate vertically about the pins or dowels of the bracket holder arms or the pins or dowels of the half ring.

(22) In another aspect, the disclosure provides a bidet washing apparatus, wherein the securing slots on the disks are substantially parallel to each other.

(23) In another aspect, the disclosure provides a bidet washing apparatus, wherein the securing slots on the disks can each independently rotate about a vertical axis from about -45 degrees to about +45 degrees.

(24) In another embodiment, the disclosure provides a bidet washing apparatus adjustably attachable to a toilet bowl for cleaning one or more body parts of a user, which includes two or more adjustable disk hinge fasteners configured for adjustably securing the bidet washing apparatus to the toilet bowl, wherein the two or more adjustable disk hinge fasteners each includes a disk having two or more openings along its perimeter, and a securing slot for securing the bidet washing apparatus to a toilet bowl; a half ring having at least two pins or dowels along its inner perimeter, each configured to engage one of the two or more openings along the perimeter of the disk; and a bracket holder having two bracket holder arms, wherein each bracket holder arm includes an internal ridge, lip or slot for securing the half ring; or a half ring having two or more openings along its inner perimeter; a disk having at least two pins or dowels along its perimeter, each configured to engage one of the two or more openings on the inner perimeter of the half ring, and a securing slot for securing the bidet washing apparatus to a toilet bowl; and a bracket holder having two bracket holder arms, wherein each bracket holder arm includes an internal ridge, lip or slot for securing the half ring; one or more water inlets configured to supply water to the bidet washing apparatus; a control unit, housing one or more valves connected to the one or more water inlets, including one or more control switches configured to operate the one or more valves, wherein the one or more valves control water flow from the one or more water inlets; and a nozzle assembly including at least one washing nozzle, connected to at least one of the one or more valves with one or more water tubes, wherein the at least one washing nozzle is positionable for directing water to the one or more body parts of the user.

(25) In one aspect, the disclosure provides a bidet washing apparatus, wherein the disks are polygon-shaped disks.

(26) In another aspect, the disclosure provides a bidet washing apparatus, wherein the disks are each independently a geometric shaped disk selected from a circle, triangle, rectangle, rhombus, square, and a trapezoid.

(27) In another aspect, the disclosure provides a bidet washing apparatus, wherein the securing slot on the disks is a U-shaped securing slot.

(28) In another aspect, the disclosure provides a bidet washing apparatus, wherein the pins or dowels along the inner perimeter of the half ring is inserted into one of the two or more openings on the disk at about 180 degrees to each other.

(29) In another aspect, the disclosure provides a bidet washing apparatus, wherein the at least two pins or dowels along the perimeter of the disk are inserted into the two or more openings along the inner perimeter of the half ring at about 180 degrees to each other.

(30) In another aspect, the disclosure provides a bidet washing apparatus, wherein each bracket holder can rotate vertically about the pins or dowels of the half ring or the pins or dowels of the disks.

(31) In another aspect, the disclosure provides a bidet washing apparatus, wherein the securing slots on the disks are substantially parallel to each other.

(32) In another aspect, the disclosure provides a bidet washing apparatus, wherein the securing slots on the disks can each independently rotate about a vertical axis from about -45 degrees to about +45 degrees.

(33) In another embodiment, the disclosure provides methods for attaching a bidet washing apparatus to a toilet bowl, which includes attaching a first end of the one or more adjustable disk hinge fasteners to securing holes in the toilet bowl; and attaching a second end of the one or more adjustable disk hinge fasteners to the bidet washing apparatus.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present disclosure, in accordance with one or more various embodiments, is described in detail with reference to the following figures. The drawings are provided for purposes of illustration only and merely depict exemplary embodiments of the disclosure. These drawings are provided to facilitate the reader's understanding of the disclosure and should not be considered limiting of the breadth, scope, or applicability of the disclosure. It should be noted that for clarity and ease of illustration these drawings are not necessarily made to scale.
- (2) FIG. 1 is a top perspective view of a bidet washing apparatus installed on a toilet bowl without a toilet seat;
- (3) FIG. 2 is a top perspective view of a bidet washing apparatus installed on a toilet bowl with the toilet seat down;
- (4) FIG. 3 is a top rear perspective view of a bidet washing apparatus having a pair of adjustable disk hinge fasteners for attaching the bidet to a toilet bowl;
- (5) FIG. 4 is an exploded top, front perspective view of an adjustable disk hinge fastener;
- (6) FIG. 5 is a cut-away of a top perspective view of the adjustable disk hinge fastener;
- (7) FIGS. 6A, 6B, 6C, 6D, 6E and 6F are top, rear perspective views of a bidet washing apparatus with the adjustable hinge fasteners at different degrees of rotation;
- (8) FIG. 7 is a top, front perspective view of a bidet washing apparatus with adjustable disk hinge fasteners;
- (9) FIG. 8 is a bottom perspective view of a bidet washing apparatus with adjustable disk hinge fasteners;
- (10) FIG. 9 is a left side perspective view of a bidet washing apparatus with adjustable disk hinge fasteners;
- (11) FIGS. 10A and 10B are right side views of a bidet washing apparatus showing the apparatus in a seated position and in a rotated upwards position, respectively;
- (12) FIG. 11 is a rear top perspective view of a bidet washing apparatus showing the apparatus in a rotated upwards position;
- (13) FIG. 12 is a top rear perspective view of a bidet washing apparatus having a pair of adjustable disk hinge fasteners for attaching the bidet to a toilet bowl;
- (14) FIG. 13 is a cut-away top perspective view of the adjustable disk hinge fastener;
- (15) FIG. 14 is a top, front perspective view of a bidet washing apparatus having adjustable disk hinge fasteners; and
- (16) FIGS. 15A, 15B, 15C, 15D, 15E, and 15F are top, rear perspective views of a bidet washing apparatus having adjustable hinge fasteners.

DETAILED DESCRIPTION OF THE INVENTION

- (17) The following description is presented to enable a person of ordinary skill in the art to make and use the embodiments described herein. Descriptions of specific devices, techniques, and applications are provided only as examples. Various modifications to the examples described herein will be readily apparent to those of ordinary skill in the art, and the general principles defined herein may be applied to other examples and applications without departing from the spirit and scope of the disclosure. Thus, the present disclosure is not intended to be limited to the examples described herein and shown but is to be accorded the scope consistent with the claims.
- (18) As used herein, the word “exemplary” means “serving as an example illustration.” Any aspect or design described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects or designs. It should be understood that the specific order or hierarchy of steps in the process disclosed herein is an example of exemplary approaches. Based upon design preferences, it is understood that the specific order or hierarchy of steps in the

processes may be rearranged while remaining within the scope of the present disclosure. Any accompanying method claims present elements of the various steps in a sample order, and are not meant to be limited to the specific order or hierarchy presented.

(19) As described herein, a “bidet washing apparatus,” or a “bidet” is a toilet attachment for cleaning the body parts of the user.

(20) As described herein, the term “water inlet” means any structure that may provide water to the bidet washing apparatus.

(21) As described herein, the term “control unit” (aka “control panel”) refers to the housing that has “control switch(es)” thereon, for controlling the various functionalities of the bidet, including but not limited to controlling the flow of water through the nozzles; controlling and/or adjusting the angle of the nozzles; and controlling the opening and/or closing of the protective shield gate.

(22) As described herein, the term “control valves” refers to the controller parts located inside the control panel housing, which controls the flow of water or other fluids from the water inlet(s) to one or more “water tubes,” by opening, closing or partially obstructing various passageways.

(23) As described herein, the term “water tubes” refers to channels that connect the control valves to a “nozzle assembly.”

(24) As described herein, the term “nozzle assembly” refers to a single nozzle or a collection of nozzles including at least one “washing nozzle.” As described herein, the term “nozzle assembly” may also have other types of nozzles such as a “self-cleaning nozzle,” which is used to clean the nozzle assembly itself, a “toilet cleaning nozzle,” which is used to clean the bidet and/or the toilet, and a “shield cleaning nozzle,” which is used for cleaning the “protective shield gate.”

(25) As described herein, the term “nozzle” refers to a device designed to eject water or other fluids as a coherent controlled spray.

(26) As described herein, the term “washing nozzle” refers to the nozzle that may be used to wash the body parts of a user.

(27) As described herein, the term “protective shield gate” refers to a structure placed at least partially in front of the nozzle assembly (e.g., between the user and the nozzle assembly) to protect the nozzle assembly from pollutants. In some embodiments, the protective shield gate may have a “hinged” edge.

(28) As described herein, the term “adjustable disk hinge fastener” is a generic term for any “rotationally adjustable disk hinge fastener,” that is used to fasten or secure a bidet washing apparatus to a toilet bowl, and which allows the apparatus to be independently moved in a horizontal direction and to pivot in a vertical direction.

(29) As described herein, the term “hinged” refers to a joint that allows the turning or pivoting of the protective shield gate, by any conventional turning or pivoting mechanism.

(30) As described herein, the term “fluidically coupled” or “fluidically connected” refers to a connection or a passageway that allows fluid to flow therethrough.

(31) The various embodiments disclosed herein provide a new, clean and hygienic bidet washing apparatus with one or more adjustable hinge fasteners, which may be attachable to an existing toilet for cleaning the body parts of the user sitting on or near the toilet.

(32) FIGS. 1 and 2 illustrate a top view of a bidet washing apparatus **100** mounted on top of a toilet bowl **105** with outlet **105a**, using securing mechanisms **110a** and **110b**. FIG. 1 illustrates a bidet washing apparatus **100** on top of the toilet bowl **105** without a toilet seat; whereas FIG. 2 illustrates a bidet washing apparatus **100** on top of the toilet bowl **105** with a toilet seat **115** in the down position.

(33) As shown in FIG. 2, the toilet seat **115** can pivot upwards while being connected to a rear portion of the toilet bowl **105** as is known in the art, wherein the toilet seat **115** partially covers the bidet washing apparatus **100** and the toilet bowl **105** when the seat is in the down position as shown. A refillable toilet tank **125** can be mounted on a rear portion of the toilet bowl **105**, in which an amount of water may be stored.

(34) In some embodiments, the bidet washing apparatus **100** may include a nozzle assembly **130**, which can include at least one washing nozzle (not shown) for washing one or more body parts, e.g. anus and genitals, of a user sitting on or near the toilet bowl **105**. In an embodiment, the bidet washing apparatus **100** may include a nozzle assembly **130** having two or more washing nozzles. However, in other embodiments a lesser or greater number of washing nozzles may be used. In some embodiments, each washing nozzle may independently spray a stream of water upwardly and inwardly towards the user. In other embodiments, the bidet washing apparatus **100** may include a nozzle assembly **130**, which has a protective shield gate **150** substantially or partially in front of it. The position of the protective shield gate **150** allows it to act as a shield between the user's body and/or water in the toilet bowl **105** and the nozzle assembly **130** and nozzles, thus protecting the nozzle assembly and nozzles from contamination or pollution during use.

(35) As shown in FIGS. **1** and **2**, the bidet washing apparatus **100** may include a control unit **140** for easy access to the user. The control unit **100** houses control switches **145a** and **145b**, which provides operating instructions to the bidet washing apparatus. The depicted example in FIGS. **1** and **2** show two control switches: **145a** and **145b**; however, one of ordinary skill in the art would realize that any number of switches may be provided for performing various operations, without departing from the scope of the present disclosure. Some examples of operating instructions include but are not limited to controlling the flow of water from a water inlet; controlling the angle of the washing nozzles; and controlling the opening and closing of a protective shield gate for protecting the washing nozzles. The type of control switches include but are not limited to knobs, dials, levers, and depressible buttons, or any conventional control mechanism known to those of skill in the art. In an embodiment, the bidet washing apparatus may have all similar control switches, wherein both control switches are knobs, dials, or levers. In other embodiments, the bidet washing apparatus may have a control switch that is a knob, and a control switch that is a dial or lever and/or vice-versa.

(36) In various embodiments, a bidet washing apparatus may be adjustably attachable to a toilet bowl through one or more adjustable disk hinge fasteners present on the bidet washing apparatus to securing holes present on a toilet bowl. In these embodiments, the adjustable hinge disk fastener includes a disk having a plurality of openings along its outer perimeter, and a securing slot for securing a bidet washing apparatus to a toilet bowl. Further, the adjustable hinge fastener includes a bracket holder having two bracket holder arms, each arm having a pin or dowel for engaging one of the plurality of openings along the perimeter of the disk.

(37) FIG. **3** illustrates a top rear view of an embodiment of a bidet washing apparatus **100** having a pair of adjustable disk hinge fasteners **200** for attaching the bidet to a toilet bowl. Also shown in this figure is the control unit **140**; control switches **145a** and **145b**; and water inlets **146a** and **146b**.

(38) As shown in FIG. **3**, each adjustable disk hinge fastener **200** includes a disk **205** having a plurality of openings **210** along the perimeter of the disk, and a securing slot **215** that can be used to connect the bidet washing apparatus **100** to a toilet bowl (not shown) using a bolt, washer and nut configuration as is known in the art (also not shown).

(39) In embodiments, the disks can each independently be of any convenient size or shape including but not limited to a flat, circular shaped disk, a polygon-shaped disk, or a geometric shaped disk selected from a circle, a triangle, a rectangle, a rhombus, a square, and a trapezoid.

(40) As shown in FIG. **3**, the disk **205** can be attached to the bidet washing apparatus **100** using bracket holders **220**, each having a pair of bracket holder arms **225**. The bracket holders **220** can be permanently or reversibly attached to the bidet washing apparatus **100** using any means known in the art, including but not limited to being built in as part of the apparatus, glued on, bolted or screwed on, and the like. In embodiments, the pair of bracket holder arms **225** can include a pin or dowel (not shown) that can engage with or be inserted into one of the plurality of openings **210** along the perimeter of the disk **205**.

(41) In other embodiments, the adjustable disk hinge fastener can include a bracket holder having

two bracket holder arms, wherein each bracket holder arm includes a plurality of openings along its inner perimeter, and a disk having a pair of pins or dowels each configured to engage one of the openings along the inner perimeter of the bracket holder arms (not shown).

(42) FIG. 4 illustrates an embodiment of an exploded top, front view of the adjustable disk hinge fastener **200**, showing the disk **205** with the openings **210** along the perimeter of the disk and securing slot **215**, bracket holder **220** with bracket holder arms **225** and the pin or dowel **230**, which can engage with or be inserted into the openings **210** on each side of the disk. This arrangement allows for easy removal of the bidet washing apparatus by allowing it to be snapped off from the brackets for better cleaning. In addition, the pivot point for rotation of the adjustable hinge fasteners, i.e., the pin or dowel bracket arm and disk configuration, allows the bracket holder and bidet washing apparatus to be vertically lifted off from a toilet bowl for a more thorough cleaning.

(43) FIG. 5 illustrates an embodiment of a cut-away of a top view of the adjustable disk hinge fastener **200**, having a disk **205** with openings **210** and securing slot **215**, bracket holder **220** and bracket holder arms **225**. As shown, the pin or dowel **230** on the bracket holder arms **225** each engage with or insert into one of the openings **210** on each side of the disk **205**, which allows the fastener (and an attached bidet washing apparatus) to pivot or rotate around the bracket holder arms and bracket holder in a vertical direction, thus facilitating cleaning of the apparatus when it is attached to a toilet bowl.

(44) FIGS. 6A, 6B, 6C, 6D, 6E and 6F illustrate embodiments of a top, rear view of a bidet washing apparatus **100** having the disks **205** with different degrees of rotation with respect to the securing slot **215**. As shown, each of the disks can be independently rotated in a horizontal direction by allowing the pin or dowel **230** to be engaged with or inserted into a desired opening **210** in the disk, thus facilitating attachment of the bidet washing apparatus to a toilet bowl. In some embodiments, the securing slots on the disks are substantially parallel to each other. In other embodiments, the securing slots on the disks can each independently rotate about a vertical axis from about -45 degrees to about +45 degrees including all degrees there between.

(45) The adjustable disk hinge fasteners can also enable a user to change the angle of the nozzle assembly (not shown) by raising or lowering the bidet washing apparatus during use. That is, the hinged operation is not only useful for being able to clean the toilet bowl or for easy installation, but it can also be used as an apparatus that makes the whole bidet (and the nozzle angle) adjustable.

(46) FIG. 7 illustrates an embodiment of a top, front view of the bidet washing apparatus **100**, which includes the adjustable disk hinge fastener **200**, disk **205** with openings **210** and securing slot **215**, bracket holder **220** and bracket holder arms **225**, and the pins or dowels (not shown), which engage with or insert into one of the plurality of openings **210** along the perimeter of the disk **205**. Also shown is the nozzle assembly **130**; washing nozzles **135**; the protective shield gate **150**; the control unit **140**; and control unit switches **145a** and **145b**.

(47) FIG. 8 illustrates an embodiment of a bottom view of the bidet washing apparatus **100**, which includes the adjustable disk hinge fastener **200**, disk **205** with openings **210** and securing slot **215**, bracket holder **220** and bracket holder arms **225**, and the pins or dowels (not shown), which engage with or insert into one of the plurality of openings **210** along the perimeter of the disk **205**. Also shown is the nozzle assembly **130**; washing nozzles **135**; the protective shield gate **150**; and the control unit **140**.

(48) FIG. 9 illustrates an embodiment of a left side view of the bidet washing apparatus **100**, which includes the adjustable disk hinge fastener **200**, disk **205** with openings **210** and securing slot **215**, bracket holder **220** and bracket holder arms **225**, and the pins or dowels (not shown), which engage with or insert into one of the plurality of openings **210** along the perimeter of the disk **205**. Also shown is the nozzle assembly **130**; washing nozzles **135**; the protective shield gate **150**; the control unit **140**; and control unit switches **145a** and **145b**.

(49) FIGS. 10A and 10B illustrates an embodiment of a right side view of the bidet washing

apparatus **100**, showing the apparatus in a seated position (FIG. **10A**) and in a rotated upwards position (FIG. **10B**), respectively. As shown, the bidet washing apparatus **100** includes the adjustable disk hinge fasteners **200**, disks **205** with openings **210** and securing slots **215**, bracket holders **220** and bracket holder arms **225**, and the pins or dowels (not shown), which engage with or insert into one of the plurality of openings **210** along the perimeter of the disk **205**. Also shown is the nozzle assembly **130**; washing nozzles **135**; the protective shield gate **150**; the control unit **140**; and control unit switches **145a** and **145b**. As shown in this figure, the bidet washing apparatus can pivot or rotate vertically upwards about the bracket arms while the disks stay in a seated position.

(50) FIG. **11** illustrates an embodiment of a rear top view of the bidet washing apparatus **100**, showing the apparatus in a rotated upwards position. As shown, the apparatus includes the adjustable disk hinge fasteners **200**, disks **205** with openings **210** and securing slots **215**, bracket holders **220** and bracket holder arms **225**, and the pins or dowels (not shown), which engage with or insert into one of the plurality of openings **210** along the perimeter of the disk **205**. Also shown is the nozzle assembly **130**; washing nozzles **135**; the protective shield gate **150**; the control unit **140**; and control unit switches **145a** and **145b**. As shown in this figure, the bidet washing apparatus can pivot or rotate upwards about the disk hinge fasteners while the disks stay in a seated position.

(51) In various other embodiments, a bidet washing apparatus can be adjustably attachable to a toilet bowl through one or more adjustable disk hinge fasteners present on the bidet washing apparatus to securing holes present on a toilet bowl. In these embodiments, the adjustable hinge disk fastener includes a disk having a securing slot for securing the bidet washing apparatus to a toilet bowl; a half ring having a plurality of openings along its outer perimeter and an internal ridge, lip or slot for securing the disk; and a bracket holder having two bracket holder arms, wherein each bracket holder arm includes a pin or a dowel configured to engage one of the openings along the outer perimeter of the half ring.

(52) FIG. **12** illustrates a top rear view of an embodiment of a bidet washing apparatus **100** having a pair of adjustable disk hinge fasteners **300** for attaching the bidet to a toilet bowl. Also shown in this figure is the control unit **140**; and control switches **145a** and **145b**.

(53) As shown in FIG. **12**, each adjustable disk hinge fastener **300** includes a disk **305** and a securing slot **310** that can be used to connect the bidet washing apparatus to a toilet bowl using a bolt, washer and nut configuration. The disk **305** can be engaged with or inserted into a rotatable half ring **315** having a plurality of openings **320** surrounding the perimeter of the half ring. The disk **305** can be attached to the half ring via an internal ridge, lip or slot **325**. The disk **305** and half ring **315** can be attached to the bidet washing apparatus via bracket holders **330** having a pair of bracket holder arms **335**.

(54) The bracket holders can be permanently or reversibly attached to the bidet washing apparatus using any means known in the art, including but not limited to being built in as part of the apparatus, glued on, screwed on and the like.

(55) In embodiments, the pair of bracket holder arms **335** can include a pin or dowel **340** that can engage with or be inserted into one of the plurality of openings **320** along the perimeter of the half ring **315**.

(56) In other embodiments, the adjustable disk hinge fastener can include a disk having a securing slot for securing the bidet washing apparatus to a toilet bowl; a bracket holder having two bracket holder arms, wherein each bracket holder arm includes a plurality of opening along an inner perimeter of the half ring and an internal ridge, lip or slot for securing the disk; and a half ring having a pair of pins or dowels each configured to engage one of the openings along the inner perimeter of the half ring (not shown).

(57) In still other embodiments, the adjustable disk hinge fastener can include a disk having an internal ridge, lip or slot for securing the half ring, and a securing slot for securing the bidet washing apparatus to a toilet bowl; a half ring having a plurality of openings along its outer

perimeter; and a bracket holder having two bracket holder arms, wherein each bracket holder arm includes a pin or a dowel configured to engage one of the openings along the outer perimeter of the half ring (not shown).

(58) FIG. 13 illustrates an embodiment of a cut-away top view of the adjustable disk hinge fastener 300, having a rotatable disk 305 and securing slot 310, rotatable half ring 315 with openings 320, bracket holder 330 with bracket holder arms 335. As shown, the pin or dowel 340 on the bracket holder arms 335 can engage with or insert into one of the openings 320 on each side of the half ring 315, which allows the fastener (and an attached bidet washing apparatus) to pivot or rotate around the bracket holder arms in a vertical direction, thus facilitating cleaning of the apparatus when it is attached to a toilet bowl.

(59) FIG. 14 illustrates an embodiment of a top, front view of the bidet washing apparatus 100, having adjustable disk hinge fasteners 300, which includes the rotatable disk 305 and securing slot 310, rotatable half ring 315 with openings 320, bracket holder 330 with bracket holder arms 335 and the pin or dowel (not shown), which engages with or inserts into the openings of the half ring. Also shown is the nozzle assembly 130; washing nozzles 135; the protective shield gate 150; the control unit 140; and control unit switches 145a and 145b.

(60) FIGS. 15A, 15B, 15C, 15D, 15E and 15F illustrate embodiments of a top, rear view of a bidet washing apparatus 100 and adjustable hinge fasteners 300, which includes the rotatable disks 305 with different degrees of rotation with respect to the securing slot 310. Also shown is the rotatable half ring 315 with openings 320, bracket holder 330 with bracket holder arms 335 and the pin or dowel (not shown), which engages with or inserts into the openings of the half ring. Also shown is the nozzle assembly 130; washing nozzles 135; the protective shield gate 150; the control unit 140; and control unit switches 145a and 145b.

(61) As shown in these figures, each of the disks can be independently rotated in a horizontal direction by rotating the disk to a desired position, thus facilitating attachment of the bidet washing apparatus to a toilet bowl. In some embodiments, the securing slots on the disks are substantially parallel to each other. In other embodiments, the securing slots on the disks can each independently rotate about a vertical axis from about -45 degrees to about +45 degrees including all degrees there between.

(62) While the inventive features have been particularly shown and described with reference to embodiments thereof, it will be understood by those in the art that the foregoing and other changes may be made therein without departing from the spirit and the scope of the disclosure. Likewise, the various diagrams may depict an example architectural or other configuration for the disclosure, which is done to aid in understanding the features and functionality that may be included in the disclosure. The disclosure is not restricted to the illustrated example architectures or configurations, but may be implemented using a variety of alternative architectures and configurations.

Additionally, although the disclosure is described above in terms of various exemplary embodiments and implementations, it should be understood that the various features and functionality described in one or more of the individual embodiments are not limited in their applicability to the particular embodiment with which they are described. They instead may be applied alone or in some combination, to one or more of the other embodiments of the disclosure, whether or not such embodiments are described, and whether or not such features are presented as being a part of a described embodiment. Thus, the breadth and scope of the present disclosure should not be limited by any of the above-described exemplary embodiments.

Claims

1. A bidet washing apparatus adjustably attachable to a toilet bowl for cleaning one or more body parts of a user, comprising: two or more adjustable disk hinge fasteners configured for adjustably securing the bidet washing apparatus to the toilet bowl, wherein the two or more adjustable disk

hinge fasteners each include: a disk having two or more openings along its perimeter, and a securing slot for securing the bidet washing apparatus to a toilet bowl; and a bracket holder having two bracket holder arms, wherein each bracket holder arm includes a pin or a dowel configured to engage one of the two or more openings along the perimeter of the disk; or a bracket holder having two bracket holder arms, wherein each bracket holder arm includes an opening; and a disk having at least two pins or dowels along its perimeter, each configured to engage one of the one or more openings on the two bracket holder arms, and a securing slot for securing the bidet washing apparatus to a toilet bowl; one or more water inlets configured to supply water to the bidet washing apparatus; a control unit, including one or more control switches configured to control water flow from the one or more water inlets; and a nozzle assembly including at least one washing nozzle, wherein the at least one washing nozzle is positionable for directing the water flow to the one or more body parts of the user.

2. The bidet washing apparatus of claim 1, wherein the disks are polygon-shaped disks.

3. The bidet washing apparatus of claim 1, wherein the disks are each independently a geometric shaped disk selected from a circle, triangle, rectangle, rhombus, square, and a trapezoid.

4. The bidet washing apparatus of claim 1, wherein the securing slot on the disks is a U-shaped securing slot.

5. The bidet washing apparatus of claim 1, wherein the pin or dowel on each of the two bracket holder arms is inserted into one of the two or more openings along the perimeter of the disk at about 180 degrees to each other.

6. The bidet washing apparatus of claim 1, wherein the at least two pins or dowels along the perimeter of the disk are inserted into the opening on the bracket holder arms at about 180 degrees to each other.

7. The bidet washing apparatus of claim 1, wherein each bracket holder can rotate vertically about the pins or dowels of the bracket holder arms or the pins or dowels of the disks.

8. The bidet washing apparatus of claim 1, wherein the securing slots on the disks are substantially parallel to each other.

9. The bidet washing apparatus of claim 1, wherein the securing slots on the disks can each independently rotate about a vertical axis from about -45 degrees to about $+45$ degrees.

10. A bidet washing apparatus adjustably attachable to a toilet bowl for cleaning one or more body parts of a user, comprising: two or more adjustable disk hinge fasteners configured for adjustably securing the bidet washing apparatus to the toilet bowl, wherein the two or more adjustable disk hinge fasteners each include: a disk having a securing slot for securing the bidet washing apparatus to a toilet bowl; a half ring having two or more openings along its perimeter, and an internal ridge, lip or slot for securing the disk; and a bracket holder having two bracket holder arms, wherein each bracket holder arm includes a pin or a dowel configured to engage one of the two or more openings along the perimeter of the half ring; or a disk having a securing slot for securing the bidet washing apparatus to a toilet bowl; a bracket holder having two bracket holder arms, wherein each bracket holder arm includes an opening; and a half ring having at least two pins or dowels along its perimeter, each configured to engage one of the one or more openings on the bracket holder arms, and an internal ridge, lip or slot for securing the disk; one or more water inlets configured to supply water to the bidet washing apparatus; a control unit, including one or more control switches configured to control water flow from the one or more water inlets; and a nozzle assembly including at least one washing nozzle, wherein the at least one washing nozzle is positionable for directing the water flow to the one or more body parts of the user.

11. The bidet washing apparatus of claim 10, wherein the disks are polygon-shaped disks.

12. The bidet washing apparatus of claim 10, wherein the disks are each independently a geometric shaped disk selected from a circle, triangle, rectangle, rhombus, square, and a trapezoid.

13. The bidet washing apparatus of claim 10, wherein the securing slot on the disks is a U-shaped securing slot.

14. The bidet washing apparatus of claim 10, wherein the pin or dowel on each of the two bracket holder arms is inserted into one of the two or more openings along the perimeter of the half ring at about 180 degrees to each other.

15. The bidet washing apparatus of claim 10, wherein the at least two pins or dowels along the perimeter of the half ring are inserted into the opening on the bracket holder arms at about 180 degrees to each other.

16. The bidet washing apparatus of claim 10, wherein each bracket holder can rotate vertically about the pins or dowels of the bracket holder arms or the pins or dowels of the half ring.

17. The bidet washing apparatus of claim 10, wherein the securing slots on the disks are substantially parallel to each other.

18. The bidet washing apparatus of claim 10, wherein the securing slots on the disks can each independently rotate about a vertical axis from about -45 degrees to about $+45$ degrees.

19. A bidet washing apparatus adjustably attachable to a toilet bowl for cleaning one or more body parts of a user, comprising: two or more adjustable disk hinge fasteners configured for adjustably securing the bidet washing apparatus to the toilet bowl, wherein the two or more adjustable disk hinge fasteners each include: a disk having two or more openings along its perimeter, and a securing slot for securing the bidet washing apparatus to a toilet bowl; a half ring having at least two pins or dowels along its inner perimeter, each configured to engage one of the two or more openings along the perimeter of the disk; and a bracket holder having two bracket holder arms, wherein each bracket holder arm includes an internal ridge, lip or slot for securing the half ring; or a half ring having two or more openings along its inner perimeter; a disk having at least two pins or dowels along its perimeter, each configured to engage one of the two or more openings on the inner perimeter of the half ring, and a securing slot for securing the bidet washing apparatus to a toilet bowl; and a bracket holder having two bracket holder arms, wherein each bracket holder arm includes an internal ridge, lip or slot for securing the half ring; one or more water inlets configured to supply water to the bidet washing apparatus; a control unit, including one or more control switches configured to control water flow from the one or more water inlets; and a nozzle assembly including at least one washing nozzle, wherein the at least one washing nozzle is positionable for directing the water flow to the one or more body parts of the user.

20. The bidet washing apparatus of claim 19, wherein the disks are polygon-shaped disks.

21. The bidet washing apparatus of claim 19, wherein the disks are each independently a geometric shaped disk selected from a circle, triangle, rectangle, rhombus, square, and a trapezoid.

22. The bidet washing apparatus of claim 19, wherein the securing slot on the disks is a U-shaped securing slot.

23. The bidet washing apparatus of claim 19, wherein the pins or dowels along the inner perimeter of the half ring is inserted into one of the two or more openings on the disk at about 180 degrees to each other.

24. The bidet washing apparatus of claim 19, wherein the at least two pins or dowels along the perimeter of the disk are inserted into the two or more openings along the inner perimeter of the half ring at about 180 degrees to each other.

25. The bidet washing apparatus of claim 19, wherein each bracket holder can rotate vertically about the pins or dowels of the half ring or the pins or dowels of the disks.

26. The bidet washing apparatus of claim 19, wherein the securing slots on the disks are substantially parallel to each other.

27. The bidet washing apparatus of claim 19, wherein the securing slots on the disks can each independently rotate about a vertical axis from about -45 degrees to about $+45$ degrees.
